The need for parallelism List

The need for parallelism

- State-of-the-art calculations require a large number of floating point operations and the time to solution must be feasible
- Calculation is run in parallel using many processor cores
- Parallel programming is vital to HPC
- Parallel programming is harder than serial programming (need a correct and efficient program)
- Why can't we just have one really fast processor?

Moore's Law

- "The number of transistors incorporated in a chip will approximately double every 24 months." - quote from Gordon Moore
- Became a target for semiconductor industry
- Progress is getting more difficult but not finished yet ...

When Apple unveils its new iPhones, expect it to make a big deal of the fact they're the first handsets in the world to be powered by a new type of chip.

This, we'll likely be told, will let owners do things like edit 4K video, enhance high-resolution photos and play graphically-intensive video games more smoothly than was possible before while using less battery power.

The "five nanometre process" involved refers to the fact that the chip's transistors have been shrunk down - the tiny on-off switches are now only about 25 atoms wide - allowing billions more to be packed in.

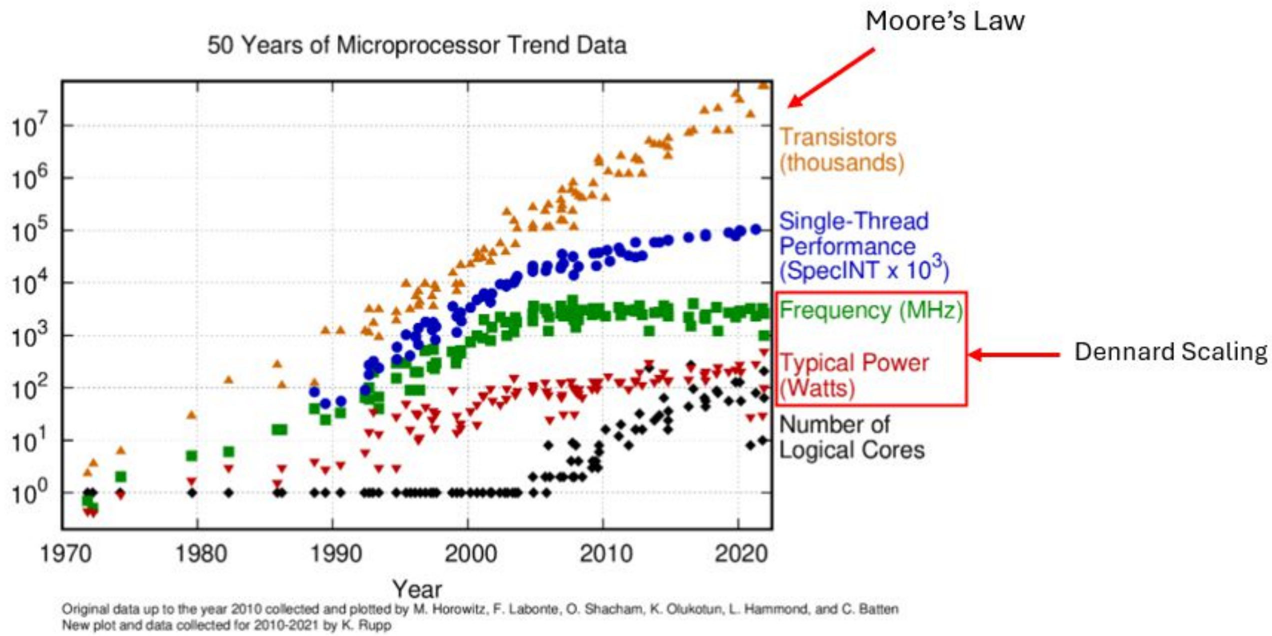
https://www.bbc.co.uk/news/technology-54510363

Dennard Scaling

- Dennard scaling is a recipe for keeping power per unit area (power density) constant as transistors were scaled to smaller sizes
- As transistors became smaller they also became faster (delay reduction) and more energy efficient (reduce threshold voltage)
- With very small features limits associated with the physics of the device (e.g. leakage current) are reached
- Dennard scaling has broken down and processor clock speeds are no longer increasing

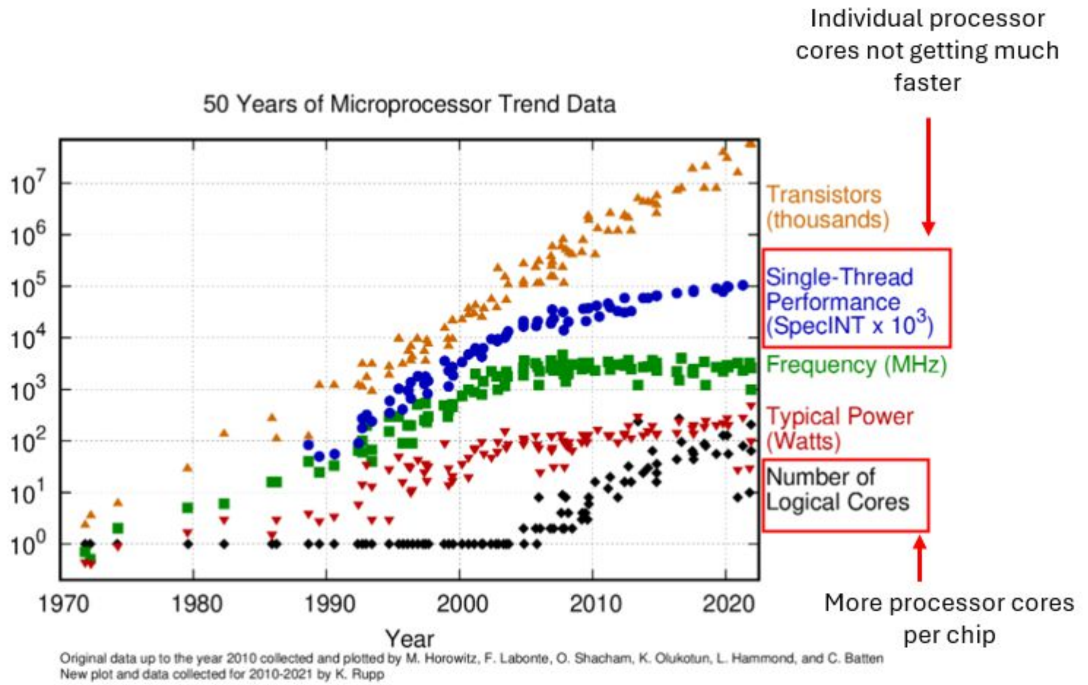
Dennard, R. et al. Design of ion-implanted MOSFETs with very small physical dimensions. IEEE Journal of Solid State Circuits SC-9, 5 (Oct. 1974), 256-268

50 Years of Microprocessor Trend Data



https://www.karlrupp.net/2018/02/42-years-of-microprocessor-trend-data/

50 Years of Microprocessor Trend Data



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Unit Summary

- The size of components on a silicon chip continues to decrease giving more components per chip (Moore's law continues)
- Clock speeds (frequencies) are not increasing any more (Dennard scaling has stopped)
- Now we get more processor cores per chip but individual core performance is not increasing significantly
- Parallelism is required to use modern multi-core processors effectively